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CPE301

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1 Project Motivation

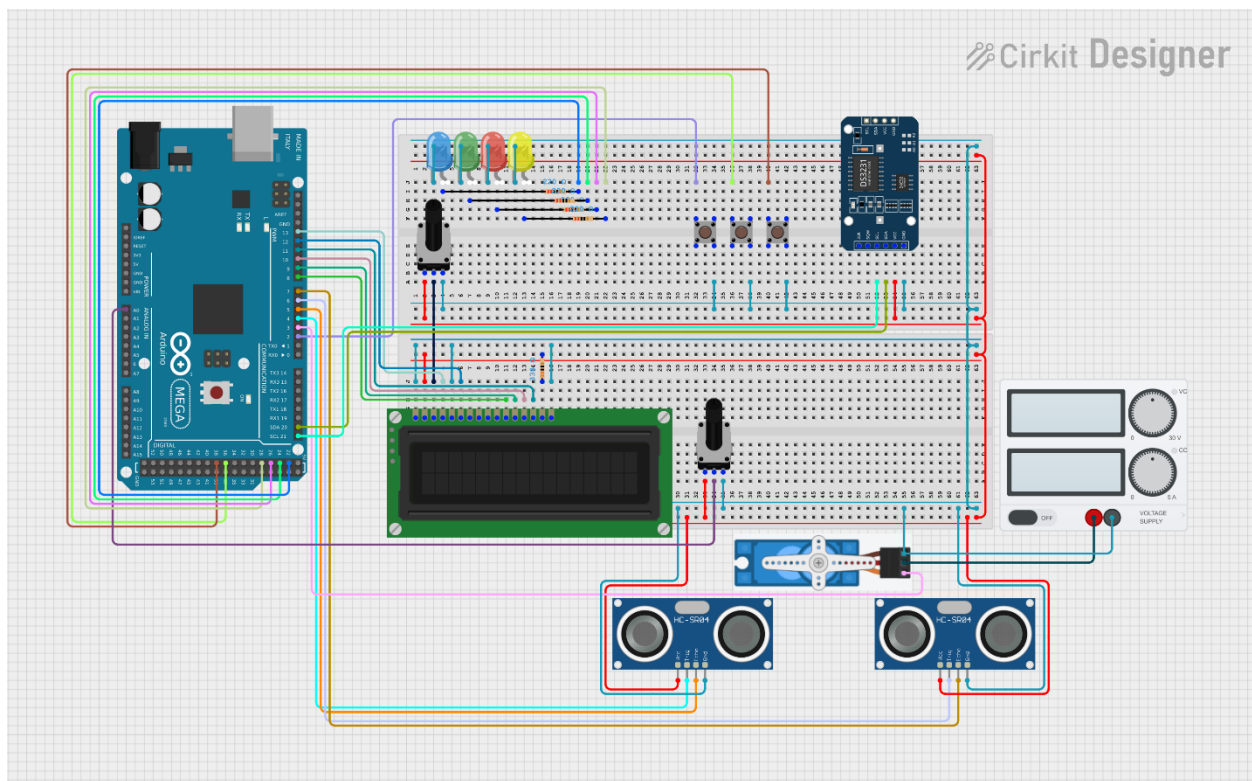
Some pet owners like to buy little forts for their pets and some like building little houses. A thing that the majority of these have in common is that they do not have doors. As it is understandable that pets do not need doors, it may still be a cute addition to the house. In this case, with enough training, they can teach their pets how to open and close their doors. This embedded system not only solves the want for pets to have privacy but also gives more insight into automatic doors for disabled people who are unable to press a button to open a door. This embedded system is a reasonable solution to this problem because it is designed to perform one specific task.

2 Proposed System Overview

The main feature of the system is simply opening and closing a door when an animal is nearby, which is activated when the ON button is pressed. The system will use an Ultrasonic Sensor and a Potentiometer as its main components. The ultrasonic sensor measures the distance of the animal, while the potentiometer allows users to decide how close the animal must be before the motor activates. One sensor will be placed near the entrance of the animal house, and another will be placed inside the house. The system will use the course-provided kit, which includes an Arduino, LEDs, LCD display, potentiometers, buttons, and a motor. The microcontroller

processes the sensor data and controls the motor and display. Four LEDs will indicate the current state of the system (OFF, IDLE, ACTIVE, or ERROR). A Real-Time Clock (RTC) will provide timestamps and log system information to the Serial Monitor. The LCD display will display the state of the system as well as sensor information. If anything goes wrong, users can press the RESET button to reset the system and return it to the IDLE state. If users do not plan to use the system for a short period of time, they can press the OFF button to shut it down.

2.1 System Schematic Diagram



2.2 Component List

Component	Quantity	Purpose	Reference
Arduino Mega 2560 / ELEGOO Mega R3	1	Microcontroller	[1]
USB Cable / Arduino Power Source	1	Powers Arduino / ELEGOO board	[1]
Breadboard	1-2	Circuit prototype	[1]
HC-SR04 Ultrasonic Sensor	2	Measures distance to detect animal presence	[1], [2]
9g Micro Servo Motor	1	Door actuator	[1]
Potentiometer (LCD Contrast)	1	Adjusts LCD contrast	[1]
Potentiometer (ADC input)	1	Adjusts distance threshold	[1]
16x2 LCD Display	1	Displays data and state info	[1]
DS3231 RTC Module	1	Timestamped logging	[1]
Push buttons	3	Start, OFF, Reset	[1]
LEDs	4	State indication	[1]
Resistors for LED (220Ω- 330Ω)	4	Limits current through LEDs	[1]
Resistor for LCD (220Ω- 330Ω)	1	Limits current for LCD backlight	[1]
Battery Pack (AA)	3 or 4	External power supply for servo	[3]

Jumper Wires	As needed	Connection between components	[1]
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[1] ELEGOO Mega R3 Start Kit

[2] Amazon listing for separately purchased HC-SR04 ultrasonic sensor

[3] Amazon listing for battery pack

3 System Behavior

When the system is OFF, LED 1 will light up, all components are inactive, and no sensor readings occur until the ON button is pressed. When the system is turned on, LED 2 will light up, indicating that the system is in the IDLE state. The distance sensors will remain idle until they detect something nearby. When the sensors detect an animal, the system will open the door by turning on the motor. There will be a delay after the animal is detected to prevent the door from closing while the animal walks into the house. To prevent the door from closing, another distance sensor will be placed inside the house near the animal. If the sensors detect that the animal is inside, the door will remain open until both sensors no longer detect anything. While the door is opening, closing, or remaining open, the system is in the ACTIVE state and LED 3 will turn on. If both sensors no longer detect anything, the door will begin closing until it is closed, then system will return to the IDLE state, indicated by LED 2. However, to make sure the door does not close if the animal immediately returns, a safety reopen condition allows the door to reopen. This may occur because the code for closing the door continues until a specific condition is met. Any errors along the way, such as invalid readings, will display error information along with a timestamp on the Serial Monitor and turn on the ERROR LED (LED

4). Pressing the RESET button clears the error and returns the system to the IDLE state. Pressing the OFF button from any state will immediately return the system to the OFF state and disable all sensors and actuators.

3.1 Module Design

Sensor Module

- Reads distance data from the ultrasonic sensors and analog input from the potentiometer, then sends this information to the Control Logic Module.
- Includes: readPot(), getDistance(), readSensors()

Control Logic Module

- Processes the sensor data and determines whether the door should open or close. It sends control signals to the Actuator Module to move the door.
- Includes: loop(), buttonISR(), sensors(), reset(), offState(), idleState(), activeState(), errorState()

Actuator Module

- Controls the motor to physically open or close the door when instructed by the Control Logic Module.
- Includes: openingDoor(), closingDoor(), safetyReopen()

Display Module

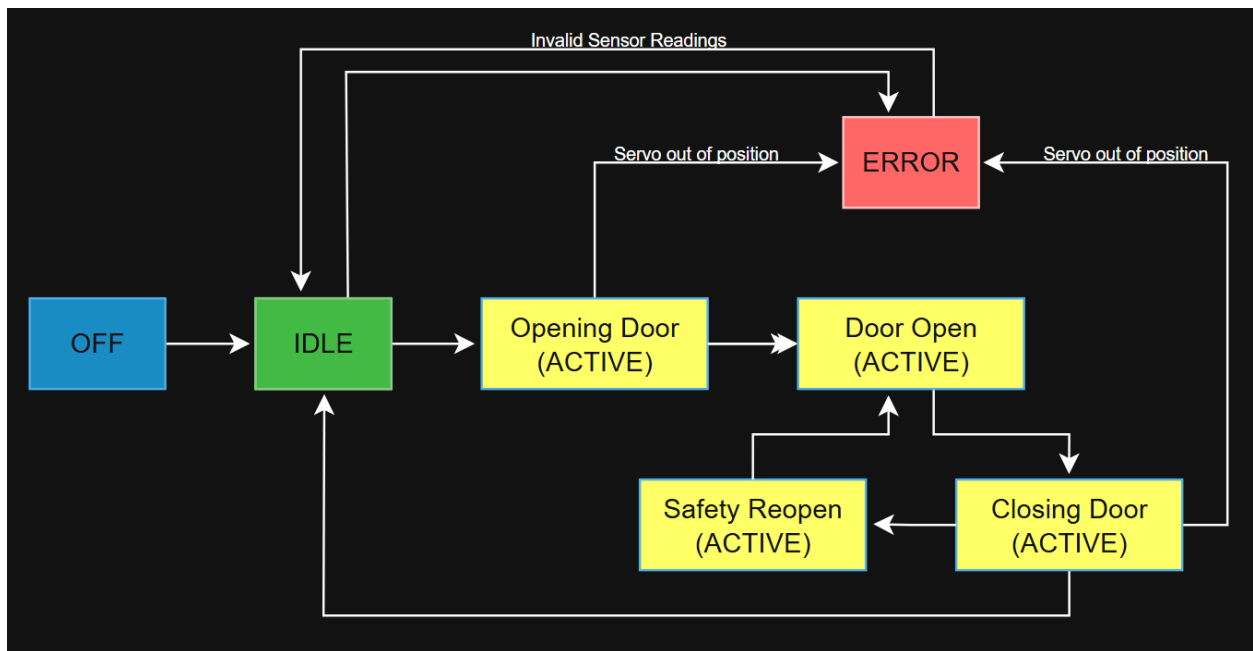
- Handles updating the LCD display with information relevant to the current status of the system.

- Includes: state(), displayData()

Communication Module

- Sends system data for debugging and monitoring purposes and works alongside the Display Module to display information with timestamps provided by the Real-Time Clock (RTC).
- Includes: logEvent()

3.2 Control Logic



OFF -> IDLE: ON button interrupt triggered

IDLE -> Opening Door (ACTIVE): Sensors detect animal

Opening Door (ACTIVE) -> Door Open (ACTIVE): Door fully opened

Door Open (ACTIVE) -> Closing Door (ACTIVE): Sensors no longer detect animal

Closing Door (ACTIVE) -> IDLE: Door fully closed

Closing Door (ACTIVE) -> Safety Reopen (ACTIVE): Animal detected while closing

Safety Reopen (ACTIVE) -> Door Open (ACTIVE): Door reopening

Opening Door (ACTIVE) -> ERROR: Door not in closed start position

Closing Door (ACTIVE) -> ERROR: Door not fully opened/closed

IDLE -> ERROR: Invalid sensor readings

ERROR -> IDLE: Reset button pressed

ANY STATE -> OFF: Off button pressed

4 Result

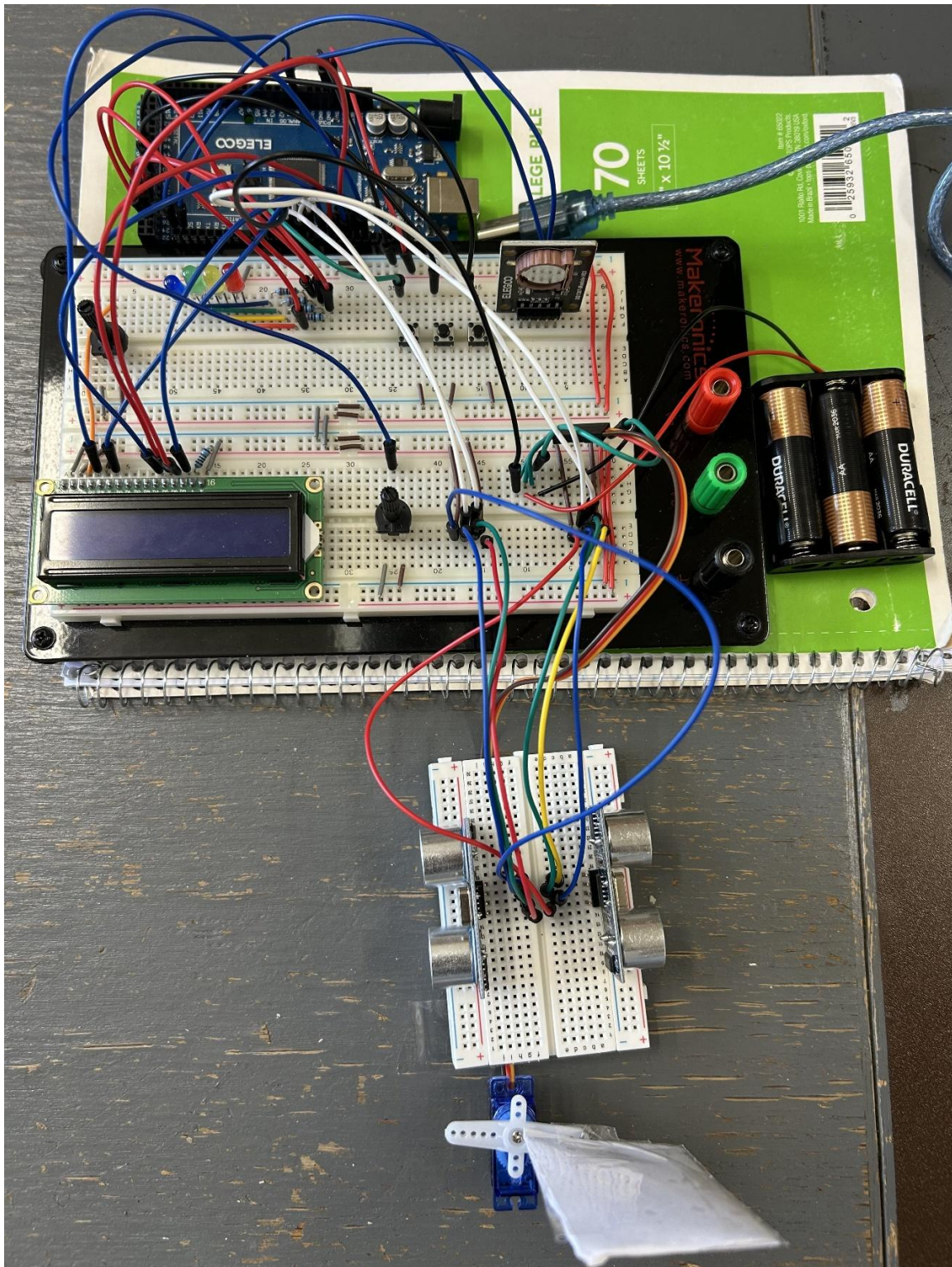


Figure 1: Full system circuit

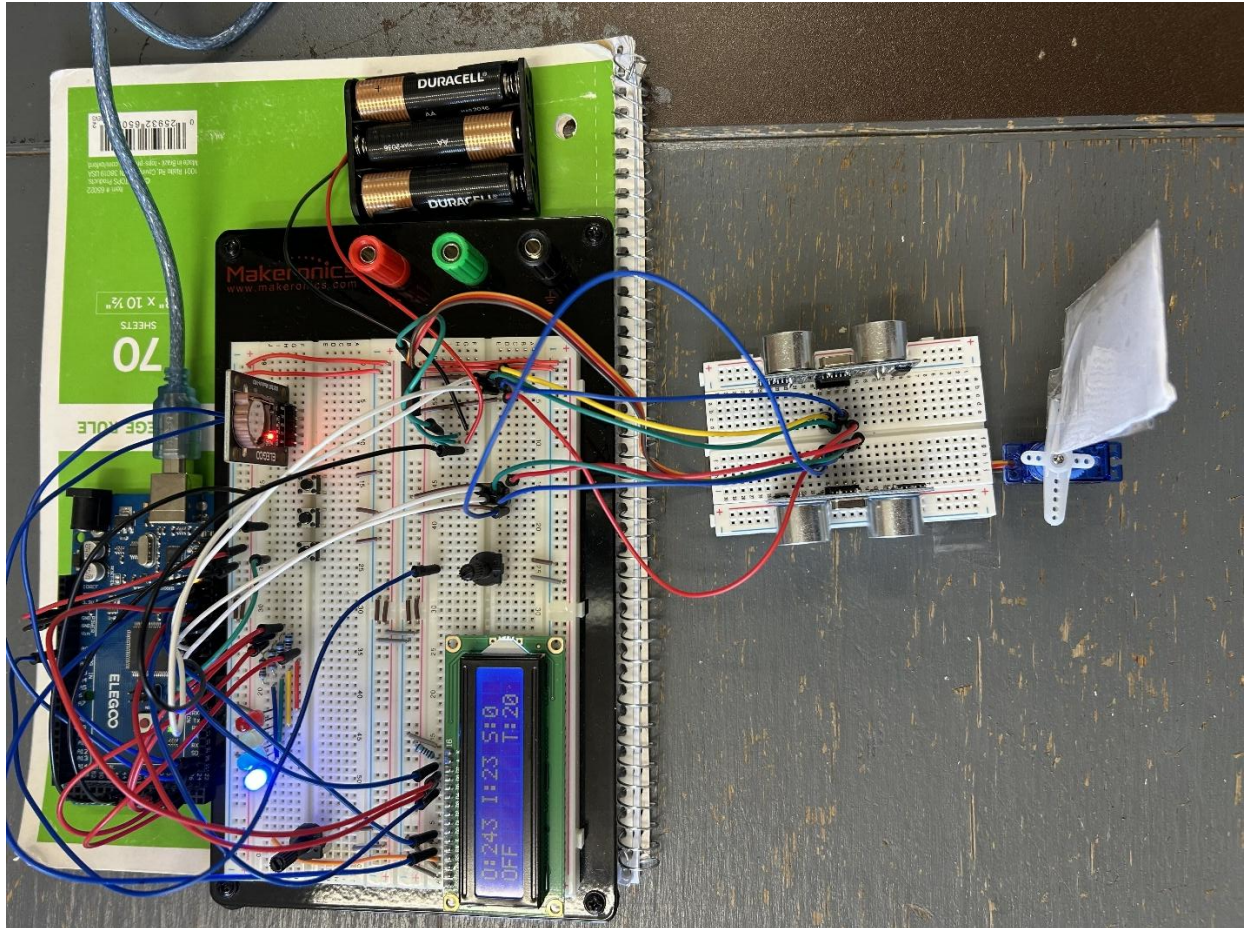


Figure 2: System in OFF state

13:13:59 - OFF state entered

Figure 3: Serial monitor logging in OFF state; program started

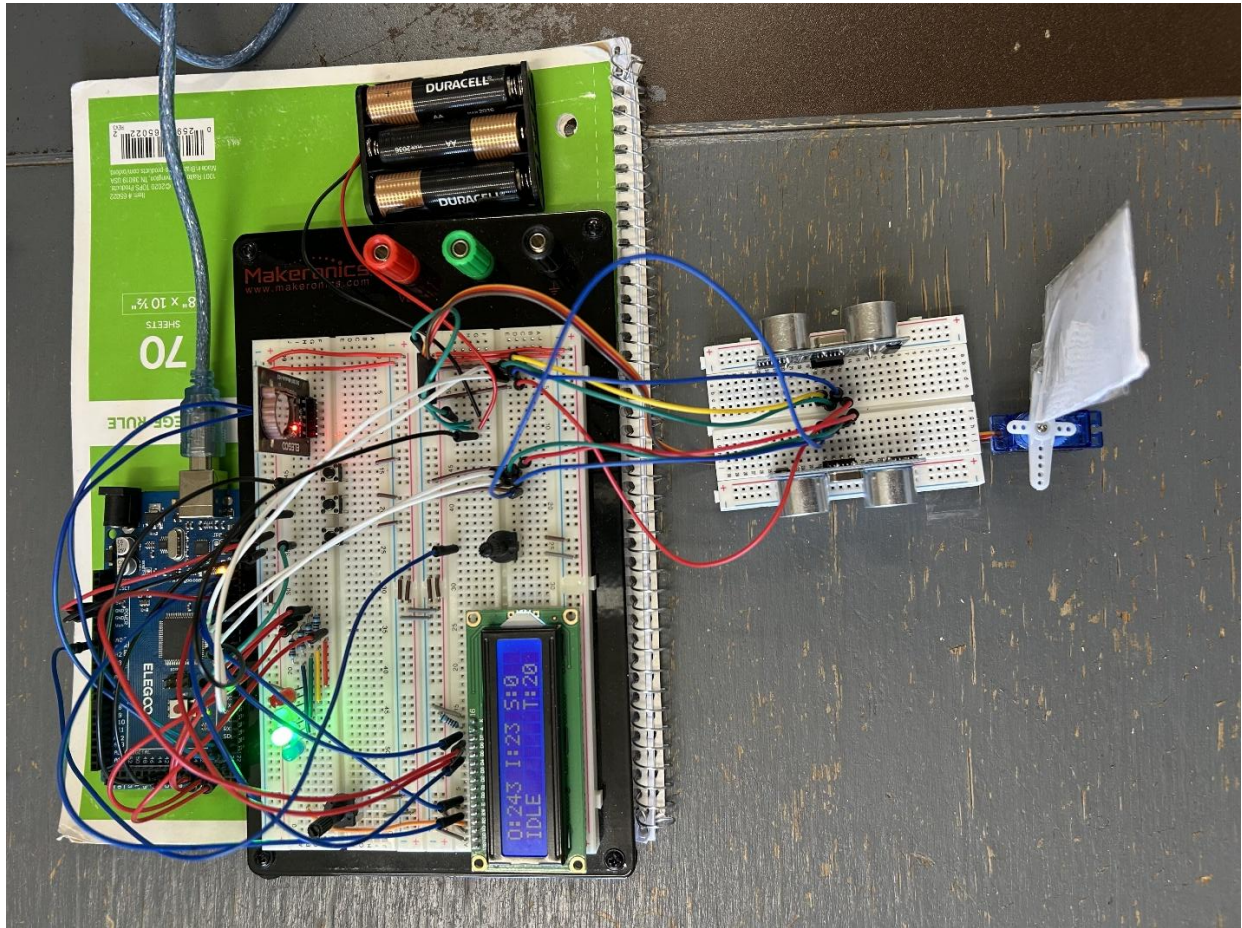


Figure 4: System in IDLE state

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13:13:59 - OFF state entered  
13:14:52 - IDLE state entered
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Figure 5: Serial monitor logging in IDLE state; start interrupt button pressed (first button from the bottom)

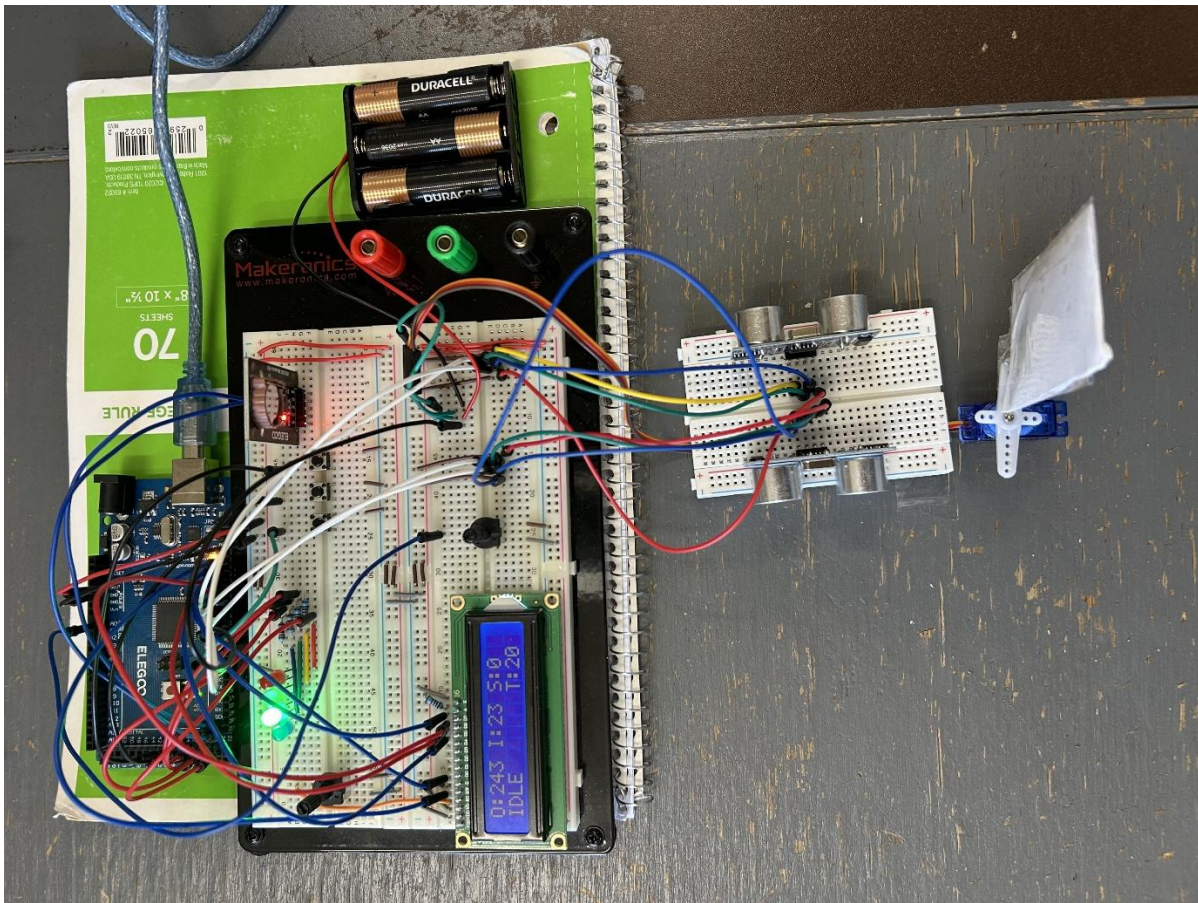


Figure 6: System in ACTIVE state

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13:13:59 - OFF state entered
13:14:52 - IDLE state entered
13:15:59 - ACTIVE state entered
13:15:59 - Door opening
13:16:02 - Door opened, monitoring animal presence
13:16:07 - Door closing
13:16:11 - Door reopening
13:16:15 - Door closing
13:16:21 - Door closed
13:16:21 - IDLE state entered

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Figure 7: Serial monitor logging in ACTIVE state; ACTIVE during opening, opened, and closing, then returned to IDLE once closed

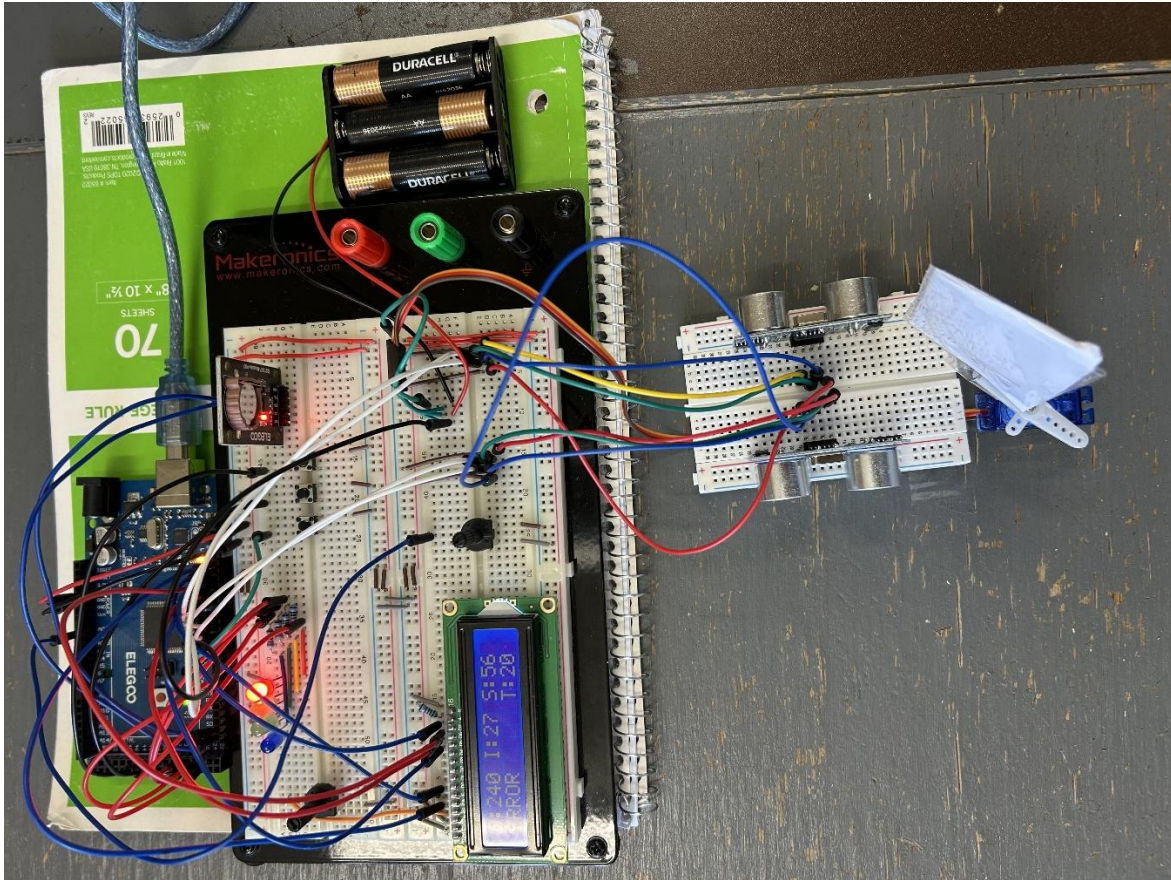


Figure 8: System in ERROR state

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13:13:59 - OFF state entered
13:14:52 - IDLE state entered
13:15:59 - ACTIVE state entered
13:15:59 - Door opening
13:16:02 - Door opened, monitoring animal presence
13:16:07 - Door closing
13:16:11 - Door reopening
13:16:15 - Door closing
13:16:21 - Door closed
13:16:21 - IDLE state entered
13:16:51 - ACTIVE state entered
13:16:51 - Door opening
13:16:53 - OFF state entered
13:16:54 - IDLE state entered
13:16:54 - ERROR state entered
13:16:54 - ERROR: Servo out of position

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Figure 9: Serial monitor logging in ERROR state; user attempted to turn off the system while the actuator was in motion

5 Environmental and Energy Impact

The embedded system uses a relatively small amount of energy to operate, as it only needs to be active when components are sensing, displaying information, or when the door is opening or closing, and the only component that moves is the servo motor. This system uses two ultrasonic sensors, an LCD display, LEDs, an RTC module, and two potentiometers, all of which are low-power components. Since this embedded system is intended for people who like to build projects for their pets, it is sustainable because many pet owners build small houses or forts for them out of recyclable materials such as foam or cardboard. They can incorporate an Arduino and a motor if they want to implement a door system for their pets. The system operates at low voltage using an Arduino. The motor will need an external power supply since the Arduino itself will most likely not be enough to safely power the components and the servo motor. Since it requires a motor to function, it does not need to be very powerful. It just needs to be strong enough to move lightweight materials such as foam or cardboard. Because the system only requires commonly available components, it is still cost-effective. The system is user-friendly because multiple LEDs indicate the system state, the LCD display shows useful system information, and the design includes OFF, ERROR, reset, and safety recovery features to make the system easier and safer to operate.

5 Conclusion

Overall, the system worked as intended. The pet door system was able to detect nearby movement using ultrasonic sensors and control the servo motor to open and close the door. The system also included multiple operating states, including OFF, IDLE, ACTIVE, and ERROR,

along with LCD output, LED indicators, reset functionality, and serial logging with RTC timestamps, which helped show the system status and improve safety and monitoring.

Some limitations of the project are that it works best in a controlled environment and depends on accurate sensor readings and correct servo positioning. The pet room must be large enough so that nothing interferes with the sensors, but also small enough for the sensors to still detect the pet's presence inside the room. If the room is too large, the animal may still be inside while being outside the sensors' detection range. The ultrasonic sensors can also return invalid readings, and the system is designed to enter an ERROR state if both sensors fail while in IDLE. The system also depends on the servo starting and remaining in the expected position, otherwise it will enter the ERROR state as well. In addition, the servo motor requires an external power supply, and the door material must remain lightweight so the motor can move it properly.